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SEAT No. _____

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SARDAR PATEL UNIVERSITY**External Examination****B.SC. (CA & IT) (INTEGRATED) – SEMESTER-I****Course No: PS01CIIT02****Subject : Computer Organization****Date : 7-11-2017, Tuesday****Time : 10:00AM to 1:00 PM****Marks : 70****Q.1 Multiple choice questions.****[10]**

- i. Which one of the following is not an input unit?

(A) Scanner	(B) Printer
(C) Keyboard	(D) None of above
- ii. Radix in the Binary Number System is _____.

(A) 0 & 1	(B) 3 & 4
(C) 5 & 6	(D) None of above
- iii. In Hexadecimal Number system, F stands for _____.

(A) 14	(B) 16
(C) 15	(D) All of above
- iv. The NAND gate has two or more input signals. If all inputs are _____, the output is high.

(A) Low	(B) High
(C) Both A & B	(D) None of these
- v. The _____ gate has two more input signals. All inputs must be same to get a low output.

(A) XOR	(B) NAND
(C) XNOR	(D) None of these
- vi. Array processor is referred as _____.

(A) SISD	(B) SIMD
(C) MIMD	(D) None of these
- vii. The world's first array processor was _____.

(A) ILLIAC VI	(B) ILLIAC IV
(C) 8086	(D) None of these
- viii. Dot matrix printer is an example of _____.

(A) Serial Printer	(B) Laser Printer
(C) Drum Printer	(D) None of these
- ix. A method for specifying an operand in memory by giving its full address, such a mode is called _____.

(A) Immediate addressing	(B) Direct addressing
(C) Register addressing	(D) none of these
- x. Which one is output device?

(A) Plotter	(B) Keyboard
(C) Mouse	(D) None of these

Q.2 Short answer (Attempt any ten questions)

[20]

- a. What is primary storage?
- b. What are the five basic operations performed by any Computer system?
- c. Convert the following :
 - i) $(185)_{10} = (?)_2$
 - ii) $(1101)_2 = (?)_{10}$
- d. Define the term Logic Gates. List the basic gates.
- e. Explain with symbol and truth table (for two inputs and three inputs) the following gates: I) AND II) NOR
- f. Draw the circuit and write the truth table for the equation: $A'B+B'C+AC'$
- g. What is vector processor??
- h. What is Multicomputer?
- i. What is secondary memory?
- j. What is Immediate addressing?
- k. Define stack addressing.
- l. What is use of scanner?

Q.3 [A] What is Number System? Explain the various number systems you know. [10]
OR

[A] Draw a block diagram of Basic Organization of a Computer System and explain the functions of the various units. [10]

Q.4 [A] Explain Hamming's code. [06]
[B] Explain the rules of Boolean algebra. [04]

OR

Q.4 [A] State and prove De Morgan's first and second theorem. [06]
[B] Explain UNICODE with example. [04]

Q.5 [A] Explain Multiprocessors. [05]
[B] Explain in detail the Von Neumann machine. [05]

OR

Q.5 [A] Explain the storage organization of optical disk. [05]
[B] Explain Superscalar Architectures. [05]

Q.6 [A] What is Direct addressing ? [03]
[B] What is addressing mode ? Explain the following addressing modes: [07]
i) Indirect addressing. ii) Indexed addressing iii) Register addressing

OR

Q.6 [A] Write a short note on Sequential Access device. [03]
[B] What is Input device ? Explain the following Input devices: [07]
i) Keyboard. ii) Scanner iii) Printer

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